

Documentation for atmgas .chpl

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Abstract

atmgas is a module for various calculations of atmospheric gas properties.

Part I

Results for mixtures of ideal gases

Note The equations used to manipulate different kinds of concentrations described here are based on a research article that I published in Portuguese ([Dias, 2021](#)), from which I also reproduce two tables. Other sources for equations are cited along the text.

The equation of state for an ideal gas is

$$pV = nR^*T, \quad (1)$$

where p is pressure, V is the volume occupied by the gas, n is the number of moles, $R^* = 8,31446261815 \text{ J mol}^{-1} \text{ K}^{-1}$ (in SI units) is the universal gas constant, and T is the thermodynamic temperature. It can also be written

$$p = n \frac{R^*}{V} T = \frac{m}{M} \frac{R^*}{V} T = \frac{m}{V} \frac{R^*}{M} T \Rightarrow p = \rho RT, \quad (2)$$

where M is molar mass and m is the mass of the gas, with

$$m = nM. \quad (3)$$

The gas constant and density are

$$R = \frac{R^*}{M}, \quad (4)$$

$$\rho = \frac{m}{V}. \quad (5)$$

Equations (1)–(5) also apply to a *mixture* of ideal gases with a few modifications. Let i be each gas in the mixture; the total number of moles is

$$n = \sum_i n_i, \quad (6)$$

where n_i is the number of moles of gas i . Assume that the gas law applies for each gas; in the partial pressure model, each gas exerts a partial pressure p_i , and all gases share a common volume V :

$$p_i V = n_i R^* T. \quad (7)$$

The total pressure is

$$p = \sum_i p_i. \quad (8)$$

Summing over (7) recovers the gas law for the mixture:

$$\begin{aligned} \sum_i (p_i V) &= \sum_i (n_i R^* T), \\ \left(\sum_i p_i \right) V &= \left(\sum_i n_i \right) R^* T, \\ pV &= n R^* T. \end{aligned}$$

By the same token, in the partial volume model each gas occupies a volume V_i under the same pressure p :

$$p V_i = n_i R^* T, \quad (9)$$

with

$$V = \sum_i V_i. \quad (10)$$

and again summing over (9) recovers the gas law for the mixture:

$$\begin{aligned} \sum_i (p V_i) &= \sum_i (n_i R^* T), \\ p \left(\sum_i V_i \right) &= \left(\sum_i n_i \right) R^* T, \\ pV &= n R^* T. \end{aligned}$$

Taking the ratio of (7) to (1) and of (9) to (1), the partial pressure fraction, the partial volume fraction, and the molar fraction are all given by

$$x_i = \frac{p_i}{p} = \frac{V_i}{V} = \frac{n_i}{n}. \quad (11)$$

The state equation for each gas can also be written

$$p_i = n_i \frac{R^*}{V} T = \frac{m_i}{M_i} \frac{R^*}{V} T = \frac{m_i}{V} \frac{R^*}{M_i} T \Rightarrow p_i = \rho_i R_i T, \quad (12)$$

in analogy with (2). The molar mass of gas i is M_i and its mass is m_i , with

$$m_i = n_i M_i. \quad (13)$$

The gas constant and density of gas i are

$$R_i = \frac{R^*}{M_i}, \quad (14)$$

$$\rho_i = \frac{m_i}{V}. \quad (15)$$

Equation (12) allows to obtain R in (2) in terms of the R_i s as follows. The total mass is

$$m = \sum_i m_i; \quad (16)$$

the gas densities also sum up:

$$\rho = \frac{m}{V} = \frac{\sum_i m_i}{V} = \sum_i \frac{m_i}{V} = \sum_i \rho_i. \quad (17)$$

This leads to the definition of the *mass fraction*,

$$c_i \equiv \frac{m_i}{m} = \frac{\rho_i}{\rho}. \quad (18)$$

Summing (12) over all components,

$$\begin{aligned} \sum_i p_i &= \sum_i \rho_i R_i T, \\ p &= \left(\sum_i \rho_i R_i \right) T \equiv \rho R T. \end{aligned} \quad (19)$$

Note that (19) defines the gas constant of the mixture. Proceeding,

$$\begin{aligned} R &= \frac{\sum_i \rho_i R_i}{\rho} \\ &= \frac{\sum_i \frac{m_i}{V} R_i}{\frac{m}{V}} \\ &= \sum_i \frac{m_i}{m} R_i = \sum_i c_i R_i. \end{aligned} \quad (20)$$

By the same token, the *mean molar mass* is

$$M = \sum_i x_i M_i = \frac{\sum_i n_i M_i}{n} = \frac{m}{n}. \quad (21)$$

Note how (21) recovers (3) explicitly for the mixture. Finally, R for the mixture can be obtained from (20):

$$\begin{aligned} R &= \frac{1}{m} \left(\sum_i m_i R_i \right) \\ &= \frac{1}{m} \left(\sum_i n_i M_i R_i \right) \\ &= \frac{1}{m} \left(\sum_i n_i R^* \right) = \frac{1}{m} \left(\sum_i n_i \right) R^* \\ &= \frac{n}{m} R^* = \frac{R^*}{M}, \end{aligned} \quad (22)$$

Table 1: Composition of a dry atmosphere according to: IRI (Iribarne and Godson, 1981), COE (COESA, 1976), WAL (Wallace and Hobbs, 2006), WIK (Wikipedia, 2020) and ADV (adopted values). After Dias (2021).

Gas	M_i (g mol ⁻¹)	x_i IRI	x_i COE	x_i WAL	x_i WIK	x_i ADV
N ₂	28.0134	0.78090000	0.78084000	0.78084000	0.78084000	0.78078700
O ₂	31.9988	0.20950000	0.20947600	0.20946000	0.20946000	0.20943200
Ar	39.948	0.00930000	0.00934000	0.00934000	0.00934000	0.00934000
CO ₂	44.0095	0.00030000	0.00031400	0.00037900	0.00040700	0.00041390
Ne	20.1797	–	0.00001818	0.00001818	0.00001818	0.00001818
He	4.002602	–	0.00000524	0.00000524	0.00000524	0.00000524
CH ₄	16.0425	–	0.00000200	0.00000170	0.00000180	0.00000170
Kr	83.798	–	0.00000114	–	0.00000114	0.00000110
H ₂	2.01588	–	0.00000055	0.00000056	0.00000055	0.00000055
N ₂ O	44.0128	–	–	0.00000031	–	0.00000033
Xe	131.293	–	0.000000087	–	–	–
soma		1.00000000	0.99999715	1.00004499	1.00007391	1.00000000

Equation (22) is the same as (4), but now obtained explicitly for the mixture.

The atmosphere is often modeled in a simplified way as a mixture of 2 gases: the main varying component, water vapor, and “dry air”. The dry air component is equivalent to the mixture of all other components. Two variables needed for the calculations involving moist air (water vapor and dry air) are the dry air mean molar mass M_d and the dry air gas constant R_d . They can be obtained from published gas concentrations in a standard atmosphere.

Table 1 shows several sources of concentrations of gases in a dry atmosphere. The standard atmosphere is defined in COESA (1976); The 1st edition of Iribarne and Godson (1981) predates it a little bit, while a relatively recent reference of dry atmosphere components is Wallace and Hobbs (2006). Non-academic references are Wikipedia (2020) and CO2.Earth (2020); the latter reports a molar fraction of 414.11 for CO₂ in 2020/03/05.

To ensure consistency, all molar mass values were obtained in NIST (2020). In table 1, the CO₂ concentration increases in time. However, for several constituents the concentrations do not change (por example, for Argon in COE, WAL e WIK): it appears that these concentrations have been reproduced in the several references without more recent measurements. Because of this, the concentrations do not always sum up to 1. In the absence of comprehensive and recent measurements, we made small adjustments to the components’ concentrations heuristically to eforce that they sum to 1 in the adopted values, in the last column.

Because different sources use slightly different values of molar masses and the universal gas constant, there are small differences between their reported values and a recalculation using the most recent values reported in NIST (2020). I calculated R_d using pressure and temperature of a standard atmosphere at sea level, $p = 101\,325$ Pa e $T = 288.15$ K (COESA, 1976). Instead of M and R for a generic mixture, I now use M_d and R_d , which are calculated as follows: (11) is used to calculate p_i ; para o cálculo da pressão parcial p_i ; (14) for the gas constants R_i ; (12) for the densities ρ_i ; (21) to obtain M_d e (20) to obtain R_d . Results are shown in table 2.

Table 2: Values of M_d and R_d reported in the references and recalculated from table 1. After Dias (2021).

Fonte	M_d (rep) (g mol ⁻¹)	M_d (calc) (g mol ⁻¹)	R_d (rep) (J kg ⁻¹ K ⁻¹)	R_d (calc) (J kg ⁻¹ K ⁻¹)
IRI	28.9640	28.9641	287.0500	287.0607
COE	28.9644	28.9644	287.0580	287.0569
WAL	28.9700	28.9667	287.0000	287.0484
WIK	–	28.9680	–	287.0436
ADV	–	28.9659	–	287.0429

In atmgas, the molar masses adopted are from column M_i of table 1, and M_d and R_d are from line ADV of table 2.

Part II

Constants and procedures

1 Constants

Gas names are used to index associative arrays. The valid gas names are "DRY", "N2", "O2", "AR", "H2O", "CO2", "NE", "HE", "KR", "XE", "CH4", "H2", "O3", "N2O" and correspond to those in table 1, except for water vapor ("H2O"). The following associative arrays are defined:

Mmol The molar masses in kg mol⁻¹.

Rgas The gas constants R_i in J kg⁻¹ K⁻¹.

For example,

```
Mmol ["N2"] = 2.80134000e-02
Rgas ["N2"] = 2.96803052e+02
```

In addition, Ru is the universal gas constant, and Rd, Rv, Rc are defined as useful aliases for Rgas["DRY"], Rgas["H2O"], Rgas["CO2"].

The full definition of Mmol is

The Mmol associative array of molar masses (kg mol⁻¹).

```
const Mmol = [
  "DRY" => 28.966413e-3,
  "N2"  => 28.013400e-3,
  "O2"  => 31.99880e-3,
  "AR"  => 39.94800e-3,
  "H2O" => 18.0153e-3,
  "CO2" => 44.00950e-3,
  "NE"  => 20.1797e-3,
  "HE"  => 4.002602e-3,
  "KR"  => 83.798e-3,
  "XE"  => 131.293e-3,
  "CH4" => 16.04250e-3,
```

```

"H2"  => 2.01588e-3,
"O3"  => 47.99820e-3,
"N2O" => 44.01280e-3
];

```

2 x2rho: convert from molar fraction to density

x2rho: converts x in molar fraction (volumetric concentration) (parts per thousand, million, billion, etc.) to density (kg m^{-3}).

```

inline proc x2rho(
  const in x: real,           // the molar fraction
  const in p: real,           // atm pressure (Pa)
  const in T: real,           // temperature (K)
  const in gas: string        // the gas name in the Mmol associative array
): real {                     // the gas density in  $\text{kg/m}^3$ 

```

3 viscair: dynamic viscosity of air

From [Montgomery \(1947\)](#).

viscair: dynamic viscosity of air (Pa s).

```

inline proc viscair(
  T: real                     // temperature (K)
): real {                     // viscosity ( $\text{Pa s}$ )

```

4 difmom: kinematic viscosity of air

difmom: kinematic viscosity of air ($\text{m}^2 \text{s}^{-1}$).

```

inline proc difmom(
  const in p: real,           // pressure (Pa)
  const in T: real,           // temperature (K)
  const in q: real            // specific humidity ( $\text{kg/kg}$ )
): real {                     // kinematic viscosity of air ( $\text{m}^2/\text{s}$ )

```

5 difheat: thermal diffusivity of air

difheat: thermal diffusivity of air ($\text{m}^2 \text{s}^{-1}$).

```

inline proc difheat(
  const in p: real,           // pressure (Pa)
  const in T: real,           // temperature (K)
  const in q: real            // specific humidity ( $\text{kg/kg}$ )
): real {                     // thermal diffusivity of air ( $\text{m}^2/\text{s}$ )

```

6 difvap: diffusivity of water vapor in air

difvap: thermal diffusivity of air ($\text{m}^2 \text{s}^{-1}$).

```

inline proc difvap(
  const in p: real,           // pressure (Pa)
  const in T: real,           // temperature (K)
  const in q: real            // specific humidity ( $\text{kg/kg}$ )
): real {                     // diffusivity of water vapor in air  $\text{m}^2/\text{s}$ 

```

7 rho2x: convert density to molar fraction

rho2x: convert density to molar fraction.

```
inline proc rho2x(  
  const in rhoi: real,      // gas density (kg/m^3)  
  const in p: real,         // pressure (Pa)  
  const in T: real,         // temperature (K)  
  const in gas: string      // the gas name in the Mmol associative array  
) : real {                  // molar fraction  
  
```

8 pressalt: atmospheric pressure as a function of altitude

pressalt: atmospheric pressure as a function of altitude in a standard atmosphere.

```
inline proc pressalt(  
  const in alt: real        // altitude (m)  
) : real {                  // pressure (Pa)  
  
```

References

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